Quiz Q1. In $\mathbb{R}^3$, let

$$\mathbf{v}_1 = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 6 \\ 9 \\ 0 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 0 \\ 0 \\ 7 \end{pmatrix}.$$

Which list is a basis of $\mathbb{R}^3$?

A. $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

B. $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$

C. $\mathbf{v}_1, \mathbf{v}_3$

D. $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4$

Quiz Q1 answer. The correct answer is D.

The vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4$ point in the three coordinate directions, so they are independent and span $\mathbb{R}^3$.

Since

$$\mathbf{v}_3 = 2\mathbf{v}_1 + 1.8\mathbf{v}_2,$$

options A and B are dependent. Option C contains only two vectors and cannot span $\mathbb{R}^3$.

Quiz Q2. A matrix $\mathbf{M} \in \mathbb{R}^{5\times 6}$. After Gaussian elimination, we have

$$\mathbf{M} \sim \begin{pmatrix} 3 & -1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

What are the rank and nullity of $\mathbf{M}$?

A. rank is 3, nullity is 3

B. rank is 2, nullity is 4

C. rank is 3, nullity is 2

D. rank is 2, nullity is 3

Quiz Q2 answer. The correct answer is A.

There are three pivots in the row-echelon form, so rank is 3.

$$\mathbf{M} \sim \begin{pmatrix} 3 & -1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

By rank-nullity, the sum of rank and nullity is 6, so nullity is also 3.